

Solution Requirements

Date	30 September 2026
Team ID	xxxxxxx
Project Name	AI StudyBuddy
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users register and log in with email and password. Passwords are hashed with bcrypt before saving, and a JWT access token is issued and verified (Bearer token) on every protected request.	High
2	Authorization levels	Two roles: student (default) and admin . Students access only their own materials; admin-only routes (/api/admin) are blocked for other roles with a 403 response.	High
3	External interfaces	Google Gemini API (gemini-3.7-flash) for AI content generation; MongoDB (via Mongoose) for data storage; React web client connected through the configured CLIENT_URL.	High

4	Transactions processing	Student uploads a .txt, .md or .pdf file; the text is extracted and sent to Gemini; the summary, flashcards, quiz and study plan are saved in the Material record. Temporary Gemini errors (503/429) are retried automatically.	High
5	Reporting	Show each student their uploaded materials with the generated summary, flashcards, quiz (question, options, answer) and study plan.	Medium
6	Business rules, etc.	Only .txt, .md and .pdf files up to 5 MB are accepted; every material belongs to one user; email must be unique; role can only be student or admin.	Medium
7	Compliance to laws or regulations	Follows basic data-privacy practice: hashed passwords, token-based access, secrets kept in environment variables. Formal GDPR-style compliance is not in the current scope.	Low
8	Other	Automatic created/updated timestamps on records, a global error handler returning JSON messages, and a health-check endpoint ("AI StudyBuddy API is running").	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Normal API requests respond quickly; AI generation depends on Gemini response time. Upload size is capped at 5 MB to keep processing fast.	Non-AI API responses < 2 seconds; AI generation < 30 seconds
2	Scalability	Stateless JWT authentication and MongoDB storage let the backend grow with more users and materials.	Supports 100+ concurrent users
3	Security & Data Privacy	Passwords hashed (bcrypt, 10 salt rounds); JWT-protected routes; CORS limited to the client URL; file-type filter on uploads; secrets stored in .env.	No plaintext passwords; 401/403 for unauthorized access

4	Reliability & Availability	Gemini calls retry up to 3 times with exponential backoff (5s, 10s, 20s) on 503/429 errors; global error handler prevents crashes.	99% uptime per month
5	Usability & Accessibility	Simple upload-and-study flow with clear error messages (e.g., "Only .txt, .md, and .pdf files are allowed").	Study content available in 3 clicks or fewer
6	Other	Maintainability: code is split into models, middleware, routes and utils. Portability: configuration through environment variables.	Modular structure; runs on any Node.js host